AI & Machine Learning Internship

Task 2: Feature Engineering, Model Optimization & Performance Comparison

Dataset: California Housing Dataset
Deliverable: Technical Evaluation Report
Script File: `AI_ML_Task2_Model_Comparison.ipynb`
Status: Complete

---

1. Executive Summary & Objective

Building upon the baseline regression pipeline established in Task 1, this task focuses on advanced model preparation and comparative performance evaluation. In professional Machine Learning workflows, models are rarely deployed after an initial iteration. Instead, data preprocessing techniques such as **Feature Scaling** are applied, followed by systematic evaluation across multiple machine learning algorithms.

The primary objective of this task is to train and evaluate three regression models—**Linear Regression**, **Ridge Regression**, and **Decision Tree Regressor**—using standardized input features from the California Housing Dataset to predict house values accurately.

2. Data Preprocessing & Methodology

To ensure robust model learning and prevent feature dominance, the following structured pipeline was executed:

- **Dataset Extraction:** Features (`X`) representing geographic and demographic housing attributes were separated from the target variable (`y` : `HousePrice`).
- **Feature Scaling (StandardScaler):** Attributes in the California Housing dataset exhibit varying scales and units (e.g., median income vs. population count). `StandardScaler` was applied to standardize features to zero mean and unit variance (z = (x - \mu) / \sigma), ensuring equal weighting during model convergence.
- **Train-Test Split:** Data was partitioned into 80% training data (`X_train` , `y_train`) and 20% testing data (`X_test` , `y_test`) with a fixed random seed (`random_state=42`) for reproducible validation.

3. Algorithms Under Evaluation

Three complementary regression algorithms were evaluated to capture both linear and non-linear patterns:

- **Linear Regression:** Serves as the primary baseline model to measure linear relationships.
- **Ridge Regression (`alpha=1.0`):** Incorporates L2 regularization to penalize large coefficients and prevent potential overfitting.
- **Decision Tree Regressor (`max_depth=5`):** A non-parametric tree-based model constrained in depth to capture non-linear feature interactions without overfitting.

4. Model Performance Comparison

Models were evaluated on unseen test data using two primary evaluation metrics: **Root Mean Squared Error (RMSE)** (measuring average prediction error magnitude in target units) and **R^2 Score** (measuring the proportion of variance explained by the model).

Model Name	RMSE (Lower is Better)	R^2 Score (Higher is Better)	Evaluation Summary
Linear Regression	0.745581	0.575788	Standard linear baseline model.
Ridge Regression	0.745554	0.575819	Slight regularization improvement over baseline.
Decision Tree Regressor BEST	**0.724234**	**0.599732**	**Highest explanatory power & lowest error.**

Key Performance Insights:

• **Decision Tree Regressor** outperformed both linear models, achieving the lowest RMSE (**0.7242**) and the highest R^2 Score (**0.5997**).
• Ridge Regression provided a minimal performance gain over unregularized Linear Regression, indicating that multicollinearity was not the primary limitation of the linear models.
• The improved score of the Decision Tree confirms the presence of non-linear interactions within the housing features (e.g., location coordinates and income interactions).

5. Visual Validation & Model Selection Justification

Visual validation via actual vs. predicted scatter plots demonstrated that predicted values for the Decision Tree model aligned more tightly along the ideal 45-degree reference line compared to the linear baselines.

Final Decision: The **Decision Tree Regressor** is selected as the optimal model for this task. It offers superior non-linear modeling capability for housing datasets while maintaining high interpretability when constrained to a maximum depth of 5.

6. Submission Deliverables Summary

- `AI_ML_Task2_Model_Comparison.ipynb` — Complete executable Jupyter Notebook containing scaling, multi-model training, evaluation loops, and scatter plots.
- `Task2_Report.pdf` — Formal 2-page documentation report.
- GitHub Repository — Hosted code and deliverables.